

Percentile Rank from Raw Data & Normal Distributions

Statistics Worksheet · Grade 9–12

Name: _____

Date: _____

Learning Objectives

- Arrange raw data from least to greatest and identify a score's position in a distribution
- Calculate percentile rank using the formula: $(\text{number of values below} \div \text{total values}) \times 100$
- Interpret percentile rank in context, including tied scores and comparing performance across different distributions

Problems

1. A class of 20 students took a quiz. The scores arranged from least to greatest are shown in the stem-and-leaf plot below. How many students scored lower than 74?

2. Using the same stem-and-leaf plot from Problem 1 (20 students), calculate the percentile rank of a student who scored 74.

$$\text{Percentile Rank} = \frac{\text{number of values below}}{\text{total values}} \times 100$$

3. A teacher recorded the following 10 test scores. Arrange them from least to greatest, then find the percentile rank of the student who scored 85.

91, 78, 85, 62, 74, 88, 85, 70, 95, 67

4. In Mr. Brown's statistics class of 25 students, the stem-and-leaf plot below shows the score distribution. What is the percentile rank of the student who scored 86?

5. The table below shows frequency data for a class of 30 students. Fill in the cumulative frequency column, then determine how many students scored below 80.

Score Range	Frequency	Cumulative Frequency
60–69	4	



Score Range	Frequency	Cumulative Frequency
70-79	9	
80-89	11	
90-99	6	

6. Jenny scored 86 on Mr. Brown's statistics test (class of 25 students). Using the stem-and-leaf plot from Problem 4, calculate her percentile rank and write a sentence interpreting what it means.

$$\text{Percentile Rank} = \frac{\text{number of values below}}{\text{total values}} \times 100$$

7. The day after her statistics test, Jenny scored 82 on Mr. Hall's chemistry test. Mr. Hall announced the class scores were approximately symmetric with a mean of 76 and a standard deviation of 4. Calculate Jenny's z-score on the chemistry test, then use the z-score to estimate her approximate percentile rank in chemistry.

$$Z = \frac{x - \mu}{\sigma}$$

8. Two students, Alex and Sam, are comparing their performances on different standardized tests. Alex scored 780 on a test with mean 700 and standard deviation 80. Sam scored 650 on a test with mean 560 and standard deviation 60. Compute each student's z-score and determine who performed better relative to their group.

$$Z = \frac{x - \mu}{\sigma}$$

9. A dataset of 15 values is listed below. A student claims that a score of 44 is at the 60th percentile. Without using a calculator, determine whether the student's claim is correct and explain your reasoning.

28, 31, 35, 38, 40, 41, 43, 44, 44, 47, 50, 53, 57, 61, 68

10. A school administered the same math exam to three classes. Class A has 30 students with mean 72 and standard deviation 6. Class B has 25 students with mean 80 and standard deviation 10. Class C has 40 students with mean 68 and standard deviation 5. Student X in Class A scored 84, Student Y in Class B scored 95, and Student Z in Class C scored 78. Rank the three students from highest to lowest percentile rank within their respective classes, and justify your answer using z-scores.



$$Z = \frac{x - \mu}{\sigma}$$



Percentile Rank from Raw Data & Normal Distributions — Answer Key

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Answer Key

1. Answer: 5 students

- List all scores below 74: 58, 63, 67, 69, 71
- Count them: that is 5 students who scored lower than 74

2. Answer: 25th percentile

- Count scores strictly below 74: 58, 63, 67, 69, 71 → 5 students
- Note: there are two students who scored 74; use only the count BELOW 74
- Percentile Rank = $(5 \div 20) \times 100 = 25$
- The student who scored 74 is at the 25th percentile

3. Answer: 50th percentile

- Arrange from least to greatest: 62, 67, 70, 74, 78, 85, 85, 88, 91, 95
- Count scores strictly below 85: 62, 67, 70, 74, 78 → 5 scores
- Percentile Rank = $(5 \div 10) \times 100 = 50$
- A score of 85 is at the 50th percentile

4. Answer: 60th percentile

- List all scores strictly below 86: 67, 72, 75, 75, 78, 79, 80, 80, 82, 83, 84, 85 — wait, count leaves below 86 in the 80s stem: 0,0,2,3,4 = 5 values; plus stems 6 and 7 = 1+5 = 6 values; total below 86 = 6 + 5 = 11... recount: stem 6: 67 (1), stem 7: 72,75,75,78,79 (5), stem 8 leaves before 6: 0,0,2,3,4 (5) → total = 15 scores below 86
- Percentile Rank = $(15 \div 25) \times 100 = 60$
- A score of 86 is at the 60th percentile

5. Answer: 13 students scored below 80

Score Range	Frequency	Cumulative Frequency
60–69	4	4
70–79	9	13
80–89	11	24
90–99	6	30

- Cumulative frequency for 60–69: 4
- Cumulative frequency for 70–79: 4 + 9 = 13
- Cumulative frequency for 80–89: 13 + 11 = 24
- Cumulative frequency for 90–99: 24 + 6 = 30



- Students who scored below 80 are in ranges 60–69 and 70–79: $4 + 9 = 13$ students

6. Answer: 60th percentile — 60% of students scored the same as or lower than Jenny

- From Problem 4, 15 students scored strictly below 86
- Percentile Rank = $(15 \div 25) \times 100 = 60$
- Interpretation: Jenny scored at the 60th percentile, meaning 60% of students in Mr. Brown's class scored the same as or lower than she did

7. Answer: $z = 1.5$, approximately the 93rd percentile

- $z = (82 - 76) \div 4 = 6 \div 4 = 1.5$
- A z-score of 1.5 on the standard normal table corresponds to a cumulative area of approximately 0.9332
- Jenny is at approximately the 93rd percentile in chemistry
- Although her raw score (82) is lower than in statistics (86), her percentile rank in chemistry (≈ 93 rd) is higher than her 60th percentile in statistics, so she performed better relative to her class in chemistry

8. Answer: Alex: $z = 1.0$; Sam: $z = 1.5$. Sam performed better relative to her group.

- Alex's z-score: $(780 - 700) \div 80 = 80 \div 80 = 1.0$
- Sam's z-score: $(650 - 560) \div 60 = 90 \div 60 = 1.5$
- A higher z-score means a higher position relative to the group
- Sam's z-score of 1.5 is greater than Alex's 1.0, so Sam performed better relative to her class

9. Answer: The claim is incorrect. The correct percentile rank of 44 is approximately 46.7 (the 47th percentile).

- The data is already arranged from least to greatest
- Count values strictly below 44: 28, 31, 35, 38, 40, 41, 43 $\rightarrow$ 7 values
- Percentile Rank = $(7 \div 15) \times 100 = 46.67 \approx 47$ th percentile
- The student's claim of the 60th percentile is incorrect; the correct rank is approximately the 47th percentile

10. Answer: Ranking (highest to lowest): Student Z ($z = 2.0$), Student X ($z = 2.0$) tied, Student Y ($z = 1.5$). Z and X tie for the highest relative rank; Y is third.

- Student X (Class A): $z = (84 - 72) \div 6 = 12 \div 6 = 2.0$
- Student Y (Class B): $z = (95 - 80) \div 10 = 15 \div 10 = 1.5$
- Student Z (Class C): $z = (78 - 68) \div 5 = 10 \div 5 = 2.0$
- A z-score of 2.0 corresponds to approximately the 97.7th percentile; a z-score of 1.5 corresponds to approximately the 93.3rd percentile
- Student X and Student Z are both at the 97.7th percentile within their classes (tied for first); Student Y is at the 93.3rd percentile (third)
- Even though Student Y's raw score of 95 is the highest among the three, she performed the worst relative to her own class

